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United States Patent	12385307
Kind Code	B2
Date of Patent	August 12, 2025
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Automatic reset device for door

Abstract

The present disclosure provides an automatic reset device for a door. The automatic reset device for a door includes a first connecting component, a second connecting component and an elastic reset member. The first connecting component is configured to be connected to a surface of a rotating door. The second connecting component is configured to be connected to a supporting surface close to a rotating shaft of the rotating door, and the second connecting component is provided with an adjustment component. The first connecting component and the second connecting component are rotatably connected to each other. A first end of the elastic reset member is connected to the first connecting component, and a second end of the elastic reset member is connected to the adjustment component. The first connecting component is stressed to rotate relative to the second connecting component.

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Appl. No.:	**18/512136**
Filed:	**November 17, 2023**

Prior Publication Data

Document Identifier	Publication Date
US 20250163744 A1	May. 22, 2025

Publication Classification

Int. Cl.: **E05F1/08** (20060101); **E05F1/12** (20060101)

U.S. Cl.:

CPC **E05F1/1215** (20130101); E05Y2900/132 (20130101)

Field of Classification Search

CPC: A47F (3/043); A47F (3/0434); A47F (2003/008); E05F (1/1215); E05F (1/12); E05F (1/1016); E05F (3/20); E05F (1/1223); E05F (1/1008); E05F (1/1025); E05F (1/1207); E05F (1/14); E05D (7/081); E05D (7/0009); E05Y (2201/21); E05Y (2201/25); E05Y (2201/484); E05Y (2201/492); E05Y (2201/496); E05Y (2201/702); E05Y (2201/704); E05Y (2201/71); E05Y (2900/20); E05Y (2900/202); E05Y (2900/30)

USPC: 16/298; 16/299; 16/301

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Background/Summary

FIELD

(1) The present disclosure relates to the technical field of door control devices, and in particular, to an automatic reset device for a door.

BACKGROUND OF THE INVENTION

(2) In daily lives, in consideration of various purposes such as theft prevention, fire protection, and convenience, people often need to use door control devices to automatically reset doors to their initial positions. The most common automatic reset device is a door closer.

(3) However, existing automatic reset devices for a door on the market cannot adjust an initial elastic deformation of a spring after mounting. In many cases, a resilience force of the automatic reset device for a door is too low, making it impossible to reset the door in a timely and stable manner. Or, if a resilience force of the automatic reset device for a door is too high, the door would be driven to be reset too fast in a door resetting process. On the one hand, a user easily gets hurt; and on the other hand, the door is easily damaged.

(4) For this purpose, the present disclosure provides an automatic reset device for a door, which can effectively solve the above problems. The automatic reset device for a door has a simple structure and can conveniently adjust the initial elastic deformation of the spring and effectively adjust a door resetting speed and force.

SUMMARY OF THE INVENTION

(5) In order to overcome the shortcomings of the prior art, the present disclosure provides an automatic reset device for a door. The automatic reset device for a door has a simple structure and can conveniently adjust the initial elastic deformation of the spring and effectively adjust a door resetting speed and force.

(6) The technical solution adopted by the present disclosure to solve the technical problem is as follows:

(7) An automatic reset device for a door, including: a first connecting component, wherein the first connecting component is configured to be connected to a surface of a rotating door; a second connecting component, wherein the second connecting component is configured to be connected to a supporting surface close to a rotating shaft of the rotating door, and the second connecting component is provided with an adjustment component; the first connecting component and the second connecting component are rotatably connected to each other; and an elastic reset member, wherein a first end of the elastic reset member is connected to the first connecting component, and a second end of the elastic reset member is connected to the adjustment component; the first connecting component is stressed to rotate relative to the second connecting component, and the elastic reset member generates an elastic deformation and causes the first connecting component to have a trend of being reset, wherein the adjustment component is configured to drive the elastic reset member to increase or decrease the elastic deformation.

(8) As the improvement of the present disclosure, the adjustment component includes a connector and an actuator; the actuator is rotatably connected to the connector; the connector is connected to the second connecting component; the actuator is connected to the second end of the elastic reset member; and the actuator drives the elastic reset member to increase or decrease the elastic deformation when rotating relative to the connector.

(9) As the improvement of the present disclosure, a first ratchet wheel part is arranged on a connection surface of the connector to the actuator, and a second ratchet wheel part is arranged on a connection surface of the actuator to the connector; and the first ratchet wheel part cooperates with the second ratchet wheel part to allow relative rotation between the connector and the actuator in a first circumferential direction and to hinder relative rotation between the connector and the actuator in a second circumferential direction.

(10) As the improvement of the present disclosure, the connector is provided with a through hole, and the actuator is provided with a coupling part; the coupling part is arranged opposite to the

through hole; the coupling part is configured to be coupled with a working end of a twisting tool; and the through hole is configured to allow the working end of the twisting tool to pass through for use.

(11) As the improvement of the present disclosure, the first connecting component is provided with a first shaft hole and a first rotating shaft; the second connecting component is provided with a second shaft hole; at least part of the adjustment component is arranged in a columnar shape to form a second rotating shaft; the second rotating shaft is rotatably inserted into the first shaft hole; the first rotating shaft is rotatably inserted into the second shaft hole; and the first rotating shaft, the second rotating shaft, the first shaft hole, and the second shaft hole are all coaxial to allow relative rotation between the first connecting component and the second connecting component.

(12) As the improvement of the present disclosure, the elastic reset member is a torsional spring; the torsional spring is inserted into the first shaft hole; a first clamping part is arranged at a lower part of the actuator; a second clamping part is arranged at an upper part of the first rotating shaft; and one end of the torsional spring is clamped to the first clamping part, and the other end of the torsional spring is clamped to the second clamping part.

(13) As the improvement of the present disclosure, the second shaft hole includes an upper shaft hole unit and a lower shaft hole unit; the upper shaft hole unit and the lower shaft hole unit are opposite and coaxial; the connector is connected to the upper shaft hole unit, so that the connector and the upper shaft hole unit form a whole; part of the first rotating shaft is inserted into and fixedly connected to one end of the first shaft hole, so that the first rotating shaft and the first shaft hole form a whole; the first rotating shaft is rotatably inserted into the lower shaft hole unit; and the second rotating shaft formed by at least part of the adjustment component is rotatably inserted into the other end of the first shaft hole.

(14) As the improvement of the present disclosure, the automatic reset device for the door further includes a first limiting column, wherein the first rotating shaft is provided with an insertion hole; the first connecting component is provided with a first through hole penetrating through a side wall of the first through hole; and the first limiting column is inserted into the insertion hole through the first through hole.

(15) As the improvement of the present disclosure, a limiting block is arranged on an outer surface of the connector; a limiting slot is arranged on an inner wall of the upper shaft hole unit; and the limiting block is inserted into the limiting slot.

(16) As the improvement of the present disclosure, the automatic reset device for the door further includes a second limiting column, wherein an insertion slot is arranged on a surface of the connector; the second connecting component is provided with a second through hole penetrating through a side wall of the upper shaft hole unit; and the second limiting column is inserted into the insertion slot through the second through hole.

(17) As the improvement of the present disclosure, the automatic reset device for the door further includes two gaskets, wherein the two gaskets are respectively located between the first shaft hole and the upper shaft hole unit and between the first shaft hole and the lower shaft hole unit.

(18) As the improvement of the present disclosure, the first connecting component includes a mounting plate; the mounting plate is provided with several first connecting holes; and the first connecting holes are configured to allow the connector to pass through for use.

(19) As the improvement of the present disclosure, a reinforcing rib is arranged on a surface of the mounting plate; and at least part of the reinforcing rib is arranged around the first connecting holes.

(20) As the improvement of the present disclosure, the first connecting component further includes a cover plate; and the cover plate is covered at one side of the mounting plate facing away from a mounting plane.

(21) As the improvement of the present disclosure, a rotating column is arranged on one side of the mounting plate close to the second connecting component; a rotating slot is arranged on one side of the cover plate close to the second connecting component; and the rotating column is rotatably

inserted into the rotating slot.
(22) As the improvement of the present disclosure, the rotating slot includes a sliding chute and a locking slot; the sliding chute is communicated to the locking slot; and the rotating column slides internally along the sliding chute and the locking slot.
(23) As the improvement of the present disclosure, a first locking hole is arranged on one side of the mounting plate facing away from the second connecting component; a second locking hole is arranged on one side of the cover plate far from the second connecting component; the first locking hole and the second locking hole are configured to allow the locking member to pass through for use, so as to fix the mounting plate with the cover plate.
(24) As the improvement of the present disclosure, the second connecting component is provided with a second connecting hole, and the second connecting hole is configured to allow the connector to pass through for use.
(25) As the improvement of the present disclosure, a mounting surface of the first connecting component and a mounting surface of the second connecting component are roughly parallel or perpendicular to each other.
(26) Beneficial effects of the present disclosure are as follows: By the arrangement of the above structure, during use, the first connecting component is connected to the surface of the rotating door, and it is generally connected to a position, close to the rotating shaft, on the surface of the rotating door. The second connecting component is then connected to the supporting surface or mounting surface such as a door frame and a wall surface, and it is generally also connected to a position, close to the rotating shaft of the rotating door, on the supporting surface or mounting surface. Further, an adjustment device is used to drive the elastic reset member to move to increase or decrease the elastic deformation of the elastic reset member, thus adjusting an elastic force of the automatic reset device for the door independently according to a need of a user, making it convenient for the use to use the automatic reset device for the door. Moreover, damage to the rotating door caused by an extremely high resilience force can be prevented, and the influence caused by an extremely low resilience force on the reset effect of the automatic reset device for the door can also be prevented.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) In order to explain the technical solutions of the embodiments of the present disclosure more clearly, the following will briefly introduce the accompanying drawings used in the embodiments. The drawings in the following description are only some embodiments of the present disclosure. Those of ordinary skill in the art can obtain other drawings based on these drawings without creative work.
(2) The present disclosure is further described below in detail in combination with the accompanying drawings and embodiments.
(3) FIG. **1** is a schematic diagram of an entire structure of the present disclosure;
(4) FIG. **2** is a schematic diagram of an entire structure of the present disclosure in another angle;
(5) FIG. **3** is a structural exploded diagram of the present disclosure in an angle;
(6) FIG. **4** is a structural exploded diagram of the present disclosure in another angle;
(7) FIG. **5** is a partially sectional view of the present disclosure along a plane;
(8) FIG. **6** is a sectional view of the present disclosure along another plane;
(9) FIG. **7** is a schematic diagram of an opened state of a cover plate of the present disclosure;
(10) FIG. **8** is an enlarged view of circle A in FIG. **7**; and
(11) FIG. **9** is a schematic diagram of another mounting state of the present disclosure.
DETAILED DESCRIPTION OF THE INVENTION

(12) Referring to FIG. **1** to FIG. **9**, an automatic reset device for a door includes: a first connecting component **10**, wherein the first connecting component **10** is configured to be connected to a surface of a rotating door; a second connecting component **20**, wherein the second connecting component **20** is configured to be connected to a supporting surface close to a rotating shaft of the rotating door, and the second connecting component **20** is provided with an adjustment component **40**; the first connecting component **10** and the second connecting component **20** are rotatably connected to each other; and an elastic reset member **30**, wherein a first end of the elastic reset member **30** is connected to the first connecting component **10**, and a second end of the elastic reset member **30** is connected to the adjustment component **40**; the first connecting component **10** is stressed to rotate relative to the second connecting component **20**, and the elastic reset member **30** generates an elastic deformation and causes the first connecting component **10** to have a trend of being reset, wherein the adjustment component **40** is configured to drive the elastic reset member **30** to increase or decrease the elastic deformation.
(13) By the arrangement of the above structure, during use, the first connecting component **10** is connected to the surface of the rotating door, and it is generally connected to a position, close to the rotating shaft, on the surface of the rotating door. The second connecting component **20** is then connected to the supporting surface or mounting surface such as a door frame and a wall surface, and it is generally also connected to a position, close to the rotating shaft of the rotating door, on the supporting surface or mounting surface. Further, an adjustment device is used to drive the elastic reset member to move to increase or decrease the elastic deformation of the elastic reset member, thus adjusting an elastic force of the automatic reset device for the door independently according to a need of a user, making it convenient for the use to use the automatic reset device for the door. Moreover, damage to the rotating door caused by an extremely high resilience force can be prevented, and the influence caused by an extremely low resilience force on the reset effect of the automatic reset device for the door can also be prevented.
(14) In this embodiment, the adjustment component **40** includes a connector **41** and an actuator **42**; the actuator **42** is rotatably connected to the connector **41**; the connector **41** is connected to the second connecting component **20**; the actuator **42** is connected to the second end of the elastic reset member **30**; and the actuator **42** drives the elastic reset member **30** to increase or decrease the elastic deformation when rotating relative to the connector **41**. By the arrangement of the above structure, the connector **41** can fixedly connect the adjustment component **40** to the second connecting component **20**, so that the adjustment component **40** can move synchronously with the second connecting component **20**, and thus change a deformation amount of the elastic reset member when the first connecting component **10** and the second connecting component **20** rotate relative to each other, thereby changing the elastic force. When the first connecting component **10** and the second connecting component **20** are relatively stationary, the elastic reset member **30** can be directly driven to increase or decrease the elastic deformation by adjusting the actuator **42**, which achieves the purpose of adjusting the magnitude of the resilience force.
(15) In this embodiment, a first ratchet wheel part **411** is arranged on a connection surface of the connector **41** to the actuator **42**, and a second ratchet wheel part **421** is arranged on a connection surface of the actuator **42** to the connector **41**; and the first ratchet wheel part **411** cooperates with the second ratchet wheel part **421** to allow relative rotation between the connector **41** and the actuator **42** in a first circumferential direction and to hinder relative rotation between the connector **41** and the actuator **42** in a second circumferential direction. By the arrangement of the above structure, the connector **41** and the actuator **42** can rotate in one way relative to each other through the first ratchet wheel part **411** and the second ratchet wheel part **421**. The second circumferential direction is the same as a direction of the resilience force of the elastic reset member. This setting can prevent an elastic reset force from driving the connector **41** and the actuator **42** to rotate relative to each other, so that the stability of the elastic reset force of a product is ensured, and the door can be reset stably. On the contrary, the first circumferential direction is opposite to the

direction of the resilience force of the elastic reset member, that is, during the relative rotation between the connector **41** and the actuator **42** in the first circumferential direction, the deformation amount of the elastic reset member can be increased, thereby increasing the resilience force of the elastic reset member.
(16) In this embodiment, the connector **41** is provided with a through hole **412**, and the actuator **42** is provided with a coupling part **422**; the coupling part **422** is arranged opposite to the through hole **412**; the coupling part **422** is configured to be coupled with a working end of a twisting tool; and the through hole **412** is configured to allow the working end of the twisting tool to pass through for use. By the arrangement of the above structure, during use, the working end of the twisting tool, for example, a tool bit of a screwdriver, passes through the through hole and is coupled to the coupling part, so that the actuator can be driven to rotate by rotating the screwdriver, and the resilience force of the elastic reset member is conveniently adjusted, making it convenient for use. The coupling part can be an inner hexagon screw hole.
(17) In this embodiment, the first connecting component **10** is provided with a first shaft hole **11** and a first rotating shaft **12**; the second connecting component **20** is provided with a second shaft hole **21**; at least part of the adjustment component **40** is arranged in a columnar shape to form a second rotating shaft; the second rotating shaft is rotatably inserted into the first shaft hole **11**; the first rotating shaft **12** is rotatably inserted into the second shaft hole **21**; and the first rotating shaft **12**, the second rotating shaft, the first shaft hole **11**, and the second shaft hole **21** are all coaxial to allow relative rotation between the first connecting component **10** and the second connecting component **20**. By the arrangement of the above structure, during use, the first rotating shaft **12**, the second rotating shaft, the first shaft hole **11**, and the second shaft hole **21** are coaxial, which can ensure that the first connecting component **10** and the second connecting component **20** can rotate around the same axis, so that the first connecting component **10** and the second connecting component **20** are stably rotatably connected to each other.
(18) In this embodiment, the elastic reset member **30** is a torsional spring; the torsional spring is inserted into the first shaft hole **11**; a first clamping part **423** is arranged at a lower part of the actuator **42**; a second clamping part **121** is arranged at an upper part of the first rotating shaft **12**; and one end of the torsional spring is clamped to the first clamping part **423**, and the other end of the torsional spring is clamped to the second clamping part **121**. By the arrangement of the above structure, use of the torsional spring as the elastic reset member is more suitable for storing elastic potential energy during rotation, and the cost is low. The torsional spring is inserted into the first shaft hole **11**, which can effectively limit the torsional spring when the torsional spring deforms and further ensure the stability of the product. Furthermore, the torsional spring can be effectively hidden, the aesthetics and safety of the product can be improved, and the user can be prevented from being pinched when the torsional spring deforms. Two ends of the torsional spring are respectively connected to the first clamping part **423** and the second clamping part **121**, which can achieve connections between the torsional spring and the first connecting component **10**, as well as between the torsional spring and the second connecting component **20**, and provide a restoring force for the first connecting component **10** and the second connecting component **20**. Moreover, when the user adjusts the actuator **42**, the torsional spring will also deform in a vertical direction to allow the second ratchet wheel part **421** to slide along a surface of the first ratchet wheel part **411**, making it convenient for the user to adjust the actuator **42**.
(19) In this embodiment, the second shaft hole **21** includes an upper shaft hole unit **211** and a lower shaft hole unit **212**; the upper shaft hole unit **211** and the lower shaft hole unit **212** are opposite and coaxial; the connector **41** is connected to the upper shaft hole unit **211**, so that the connector **41** and the upper shaft hole unit **211** form a whole; part of the first rotating shaft **12** is inserted into and fixedly connected to one end of the first shaft hole **11**, so that the first rotating shaft **12** and the first shaft hole **11** form a whole; the first rotating shaft **12** is rotatably inserted into the lower shaft hole unit **212**; and the second rotating shaft formed by at least part of the adjustment component **40** is

rotatably inserted into the other end of the first shaft hole **11**. By the arrangement of the above structure, the upper shaft hole unit **211** and the lower shaft hole unit **212** are respectively configured to cooperate with the first rotating shaft and the second rotating shaft, which can ensure that both ends of the component where the first shaft hole is located can be stressed to ensure that the first connecting component **10** and the second connecting component **20** are stably rotatably connected to each other.
(20) In this embodiment, the automatic reset device for the door further includes a first limiting column **50**; the first rotating shaft **12** is provided with an insertion hole **122**; the first connecting component **10** is provided with a first through hole **111** penetrating through a side wall of the first through hole **11**; and the first limiting column **50** is inserted into the insertion hole **122** through the first through hole **111**. By the arrangement of the above structure, the first limiting column **50** is inserted into the insertion hole **122** through the first through hole **111**, which can connect the component with the first rotating shaft **12** to the component with the first shaft hole **11** to form a whole. The structure is simple, and the mounting is convenient. Preferably, a thread is arranged on an inner wall of the first through hole **111**. The thread is matched with a thread at a tail end of the first limiting column **50** to stably connect the first limiting column **50** to the first through hole **111** and the insertion hole **122**, so that separation of the first limiting column **50** is prevented, and the stability of the product is ensured.
(21) In this embodiment, a limiting block **413** is arranged on an outer surface of the connector **41**; a limiting slot **2111** is arranged on an inner wall of the upper shaft hole unit **211**; and the limiting block **413** is inserted into the limiting slot **2111**. By the cooperation between the limiting block **413** and the limiting slot **2111**, movement of the connector **41** in a circumferential direction of the upper shaft hole unit **211** can be prevented. When the user adjusts the torsional spring through the actuator, fixed connection between the connector and the second connector **20** can be ensured, and the stability of the product can be improved.
(22) In this embodiment, the automatic reset device for the door further includes a second limiting column **60**; an insertion slot **414** is arranged on a surface of the connector **41**; the second connecting component **20** is provided with a second through hole **2112** penetrating through a side wall of the upper shaft hole unit **211**; and the second limiting column **60** is inserted into the insertion slot **414** through the second through hole **2112**. By the arrangement of the above structure, during use, the second limiting column **60** is inserted into the insertion slot **414** through the second through hole **2112**, which can further fix the connector **41** and the second connecting component **20** to form a whole, so as to prevent movement of the connector **41** in an axial direction of the upper shaft hole unit **211**, and improve the stability of the product.
(23) In this embodiment, the automatic reset device for the door further includes two gaskets **70**; the two gaskets **70** are respectively located between the first shaft hole **11** and the upper shaft hole unit **211** and between the first shaft hole **11** and the lower shaft hole unit **212**. By the arrangement of the above structure, the gaskets **70** can effectively protect the components with the rotating shafts and the shaft holes. Friction is reduced, so that the relative rotation between the first connecting component **10** and the second connecting component **20** is smoother, and the service life of the product can be prolonged.
(24) In this embodiment, the first connecting component **10** includes a mounting plate **13**; the mounting plate **13** is provided with several first connecting holes **131**; and the first connecting holes **131** are configured to allow the connector to pass through for use. By the arrangement of the above structure, during use, connectors such as screws are used to pass through the first connecting holes **131** to fix and connect the first connecting component to a surface of the door, which can achieve convenient mounting and use of the product.
(25) In this embodiment, a reinforcing rib **132** is arranged on a surface of the mounting plate **13**; and at least part of the reinforcing rib **132** is arranged around the first connecting holes **131**. By the arrangement of the above structure, the reinforcing rib **132** can improve the strength of the product,

while part of the reinforcing rib **132** is arranged around the first connecting holes **131**, which can improve the strength of the product at the first connecting holes **131** and further ensure the stability of the product.

(26) In this embodiment, the first connecting component **10** further includes a cover plate **14**; and the cover plate **14** is covered at one side of the mounting plate **13** facing away from a mounting plane. By the arrangement of the above structure, the cover plate **14** can be covered at the mounting plate **13**. This can cover the mounting plate and the connector such as a screw on the mounting plate, and maintain the stability and aesthetics of the product.

(27) In this embodiment, a rotating column **133** is arranged on one side of the mounting plate **13** close to the second connecting component **20**; a rotating slot **141** is arranged on one side of the cover plate **14** close to the second connecting component **20**; and the rotating column **133** is rotatably inserted into the rotating slot **141**. By the arrangement of the above structure, the rotatable connection between the mounting plate **13** and the cover plate **14** can be achieved through the cooperation between the rotating column **133** and the rotating slot **141** during use, which further facilitates the user to open the cover plate **14** during mounting and facilitates use.

(28) In this embodiment, the rotating slot **141** includes a sliding chute **1411** and a locking slot **1412**; the sliding chute **1411** is communicated to the locking slot **1412**; and the rotating column **133** can slide internally along the sliding chute **1411** and the locking slot **1412**. By the arrangement of the above structure, when the cover plate **14** is opened, the user can move the cover plate **14** to slide the rotating column **133** from the locking slot **1412** to a tail end of the sliding chute **1411**, and reserve a space between the cover plate **14** and the mounting plate **13** to allow the relative rotation between the cover plate **14** and the mounting plate **13**. When the cover plate **14** is closed, the user can move the cover plate **14** to close the mounting plate **13**, and can then slide the rotating column **133** from the tail end of the sliding chute **1411** to the locking slot **1412**, so that the cover plate **14** is in close fit with the mounting plate **13** to improve the leakproofness between the cover plate **14** and the mounting plate **13**.

(29) In this embodiment, a first locking hole **134** is arranged on one side of the mounting plate **13** facing away from the second connecting component **20**; a second locking hole **142** is arranged on one side of the cover plate **14** far from the second connecting component **20**; the first locking hole **134** and the second locking hole **142** are configured to allow the locking member to pass through for use, so as to fix the mounting plate **13** with the cover plate **14**. By the arrangement of the above structure, when the cover plate **14** is covered at the mounting plate **13**, locking members such as screws are used to pass through the first locking hole **134** and the second locking hole **142** to fixedly connect the cover plate **14** to the mounting plate **13**, which further improves the stability of the product and prevents the cover plate **14** from being separated from the mounting plate **13**.

(30) In this embodiment, the second connecting component **20** is provided with a second connecting hole **22**, and the second connecting hole **22** is configured to allow the connector to pass through for use. By the arrangement of the above structure, during use, the user can thread the connector such as a screw to through the second connecting hole **22** and mount the second connecting component **20** on the mounting plane such as a door frame and a wall surface.

(31) In this embodiment, a mounting surface of the first connecting component **10** and a mounting surface of the second connecting component **20** are roughly parallel or perpendicular to each other. By the arrangement of the above structure, the automatic reset device for the door can be adapted to different mounting scenarios. For example, in a case that the mounting plane such as a door frame and a wall surface is parallel or perpendicular to the surface of the rotating door, the automatic reset device for the door can be used.

(32) As described above, one or more embodiments are provided in conjunction with the detailed description, The specific implementation of the present disclosure is not confirmed to be limited to that the description is similar to or similar to the method, the structure and the like of the present

disclosure, or a plurality of technical deductions or substitutions are made on the premise of the conception of the present disclosure to be regarded as the protection of the present disclosure.

Claims

1. An automatic reset device for a door, comprising: a first connecting component, wherein the first connecting component is configured to be connected to a surface of a rotating door; a second connecting component rotatably connected to the first connecting component, wherein the second connecting component is provided with an adjustment component; and an elastic reset member, wherein a first end of the elastic reset member is connected to the first connecting component, and a second end of the elastic reset member is connected to the adjustment component; the first connecting component is stressed to rotate relative to the second connecting component, and the elastic reset member generates an elastic deformation and causes the first connecting component to have a resetting trend, wherein the adjustment component is configured to drive the elastic reset member to increase or decrease the elastic deformation, and the adjustment component comprises a connector and an actuator, the actuator is rotatably connected to the connector, the connector is connected to the second connecting component, the actuator is connected to the second end of the elastic reset member, and the actuator drives the elastic reset member to increase or decrease the elastic deformation when rotating relative to the connector; wherein a first ratchet wheel part is arranged on a connection surface of the connector to the actuator, and a second ratchet wheel part is arranged on a connection surface of the actuator to the connector; and the first ratchet wheel part cooperates with the second ratchet wheel part to allow relative rotation between the connector and the actuator in a first circumferential direction and to hinder relative rotation between the connector and the actuator in a second circumferential direction; wherein the connector is provided with a through hole, and the actuator is provided with a coupling part; the coupling part is arranged opposite to the through hole; the coupling part is configured to be coupled with a working end of a twisting tool; and the through hole is configured to allow the working end of the twisting tool to pass through for use; and wherein the first connecting component is provided with a first shaft hole and a first rotating shaft; the second connecting component is provided with a second shaft hole; at least part of the adjustment component is arranged in a columnar shape to form a second rotating shaft; the second rotating shaft is rotatably inserted into the first shaft hole; the first rotating shaft is rotatably inserted into the second shaft hole; and the first rotating shaft, the second rotating shaft, the first shaft hole, and the second shaft hole are all coaxial to allow relative rotation between the first connecting component and the second connecting component.
2. The automatic reset device for the door according to claim 1, wherein the elastic reset member is a torsional spring; the torsional spring is inserted into the first shaft hole; a first clamping part is arranged at a lower part of the actuator; a second clamping part is arranged at an upper part of the first rotating shaft; and one end of the torsional spring is clamped to the first clamping part, and the other end of the torsional spring is clamped to the second clamping part.
3. The automatic reset device for the door according to claim 1, wherein the second shaft hole comprises an upper shaft hole unit and a lower shaft hole unit; the upper shaft hole unit and the lower shaft hole unit are opposite and coaxial; the connector is connected to the upper shaft hole unit, so that the connector and the upper shaft hole unit form a whole; part of the first rotating shaft is inserted into and fixedly connected to one end of the first shaft hole, so that the first rotating shaft and the first shaft hole form a whole; the first rotating shaft is rotatably inserted into the lower shaft hole unit; and the second rotating shaft formed by at least part of the adjustment component is rotatably inserted into the other end of the first shaft hole.
4. The automatic reset device for the door according to claim 3, further comprising a first limiting column, wherein the first rotating shaft is provided with an insertion hole; the first connecting component is provided with a first through hole penetrating through a side wall of the first shaft

hole; and the first limiting column is inserted into the insertion hole through the first through hole.
5. The automatic reset device for the door according to claim 3, wherein a limiting block is arranged on an outer surface of the connector; a limiting slot is arranged on an inner wall of the upper shaft hole unit; and the limiting block is inserted into the limiting slot.
6. The automatic reset device for the door according to claim 3, further comprising a second limiting column, wherein an insertion slot is arranged on a surface of the connector; the second connecting component is provided with a second through hole penetrating through a side wall of the upper shaft hole unit; and the second limiting column is inserted into the insertion slot through the second through hole.
7. The automatic reset device for the door according to claim 3, further comprising two gaskets, wherein the two gaskets are respectively located between the first shaft hole and the upper shaft hole unit and between the first shaft hole and the lower shaft hole unit.
8. The automatic reset device for the door according to claim 3, wherein the first connecting component comprises a mounting plate; the mounting plate is provided with several screw connecting holes.
9. The automatic reset device for the door according to claim 8, wherein a reinforcing rib is arranged on a surface of the mounting plate; and at least part of the reinforcing rib is arranged around the first connecting holes.
10. The automatic reset device for the door according to claim 8, wherein the first connecting component further comprises a cover plate; and the cover plate is covered at one side of the mounting plate facing away from a mounting plane.
11. The automatic reset device for the door according to claim 10, wherein a rotating column is arranged on one side of the mounting plate close to the second connecting component; a rotating slot is arranged on one side of the cover plate close to the second connecting component; and the rotating column is rotatably inserted into the rotating slot.
12. The automatic reset device for the door according to claim 11, wherein the rotating slot comprises a sliding chute and a locking slot; the sliding chute is communicated to the locking slot; and the rotating column slides internally along the sliding chute and the locking slot.
13. The automatic reset device for the door according to claim 11, wherein a first locking hole is arranged on one side of the mounting plate facing away from the second connecting component; a second locking hole is arranged on one side of the cover plate far from the second connecting component; the first locking hole and the second locking hole are configured to allow a locking member to pass through for use, so as to fix the mounting plate with the cover plate.
14. The automatic reset device for the door according to claim 1, wherein the second connecting component is provided with a second connecting hole, and the second connecting hole is configured to allow the connector to pass through for use.